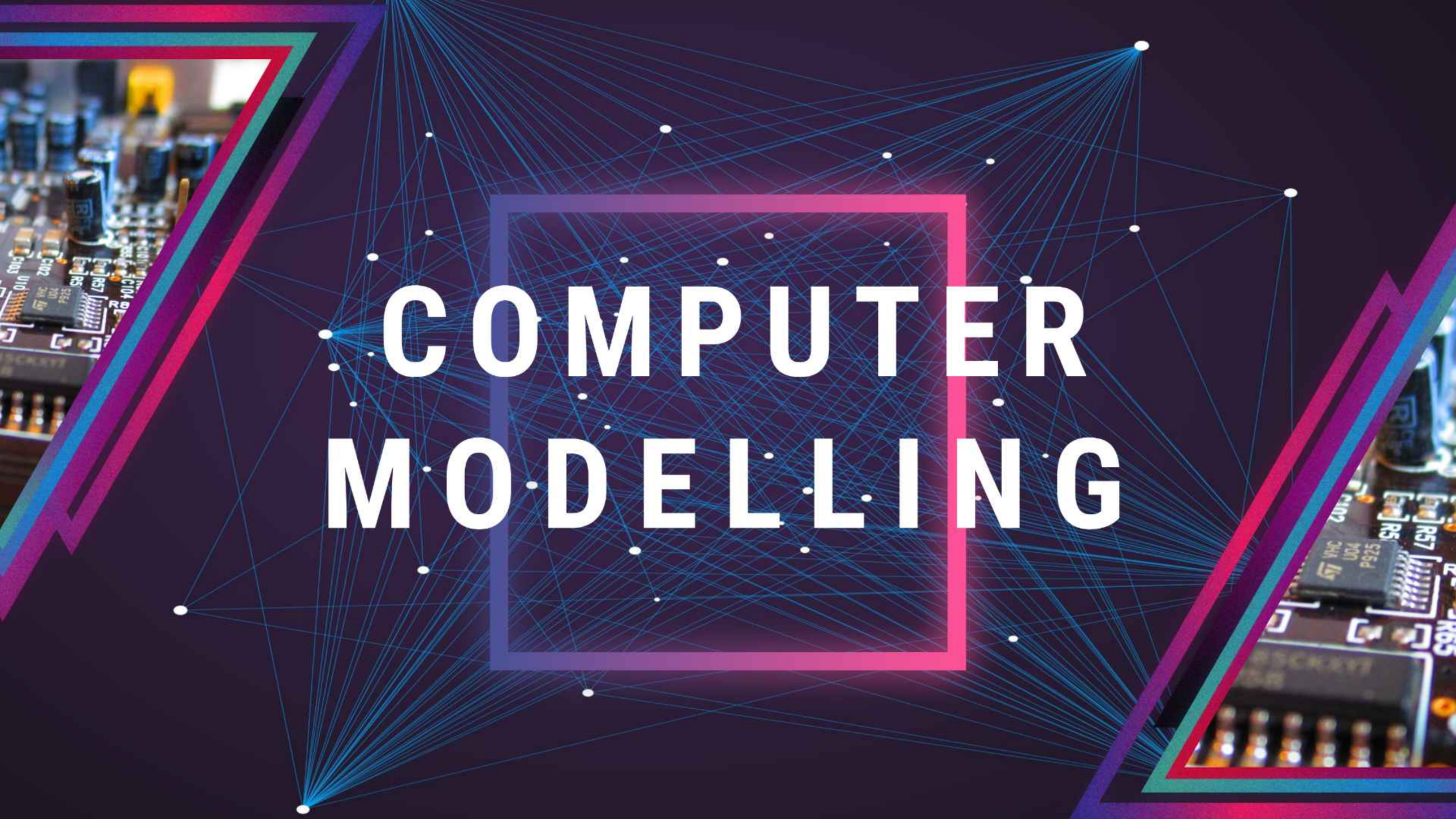

COMPUTER
MODELLING

SIMULATION

- It is the creation of a model of a real system in order to study the behaviour of the system. It is computer-generated and is based on mathematical representations.



ADVANTAGES OF USING MODELS

- Less expensive than to build the real thing
- It allows you to try out various different scenarios in advance
- Often faster to use computer model than do the real thing.



DISADVANTAGES OF USING MODELS

- A model is only as good as the programming or the data entered
- People's reactions to the results of a simulation may not be positive; they may not trust the result it produces.



EXAMPLES WHERE COMPUTER MODELLING IS USED

- Personal finance - Excel Application
- Bridge and building design
- Flood water management
- Traffic management
- Weather forcasting



PERSONAL FINANCE

- We will use the spreadsheet to model the sales of school supplies business in school
- It is part of computer modeling because it uses data, formulas, and functions to represent and simulate real-world financial scenarios.



COMPANY NAME

Street Address

City, State 12345

Phone: (000) 000-0000

REVENUE	2025	2026
Gross Sales	$ 200.00	$ 300.00
Less Sales Returns and Allowances	$ 10.00	$ 10.00
NET SALES	**$ 190.00**	**$ 290.00**

COST OF SALES	2025	2026
Beginning inventory	$ 300.00	$ 310.00
Plus Goods Purchased or Manufactured	$ 100.00	$ 120.00
TOTAL GOODS AVAILABLE	$ 400.00	$ 430.00
Less ending inventory	$ 200.00	$ 250.00
TOTAL COST OF GOODS SOLD (COGS)	**$ 200.00**	**$ 180.00**

GROSS PROFIT (LOSS)	2025	2026
	$ (10.00)	**$ 110.00**

OPERATING EXPENSES	2025	2026
SELLING		
Salaries and Wages	$ 30.00	$ 35.00
Commissions	$ 10.00	$ 20.00
Advertising	$ 10.00	$ 20.00
Depreciation	$ 20.00	$ 30.00
Other (i.e. Professional Fees)	$ 5.00	$ 6.00
TOTAL SELLING EXPENSES	**$ 75.00**	**$ 111.00**

Smartsheet Inc. ©2025



COMPANY NAME

Street Address

City, State 12345

Phone: (000) 000-0000

DATE PREPARED	01/01/2025
START YEAR	2025
END YEAR	2029

REVENUE	2025	2026	2027	2028	2029
Gross Sales	$ 200.00	$ 300.00	$ 400.00	$ 500.00	$ 600.00
Less Sales Returns and Allowances	$ 10.00	$ 10.00	$ 10.00	$ 10.00	$ 12.00
NET SALES	$ 190.00	$ 290.00	$ 390.00	$ 490.00	$ 588.00

COST OF SALES	2025	2026	2027	2028	2029
Beginning inventory	$ 300.00	$ 310.00	$ 320.00	$ 330.00	$ 340.00
Plus Goods Purchased or Manufactured	$ 100.00	$ 120.00	$ 140.00	$ 160.00	$ 170.00
TOTAL GOODS AVAILABLE	$ 400.00	$ 430.00	$ 460.00	$ 490.00	$ 510.00
Less ending inventory	$ 200.00	$ 250.00	$ 300.00	$ 350.00	$ 350.00
TOTAL COST OF GOODS SOLD (COGS)	$ 200.00	$ 180.00	$ 160.00	$ 140.00	$ 160.00

GROSS PROFIT (LOSS)	2025	2026	2027	2028	2029
	$ (10.00)	$ 110.00	$ 230.00	$ 350.00	$ 428.00

OPERATING EXPENSES	2025	2026	2027	2028	2029
SELLING					
Salaries and Wages	$ 30.00	$ 35.00	$ 40.00	$ 45.00	$ 50.00
Commissions	$ 10.00	$ 20.00	$ 30.00	$ 40.00	$ 40.00
Advertising	$ 10.00	$ 20.00	$ 30.00	$ 40.00	$ 40.00
Depreciation	$ 20.00	$ 30.00	$ 40.00	$ 50.00	$ 50.00
Other (i.e. Professional Fees)	$ 5.00	$ 6.00	$ 7.00	$ 8.00	$ 8.00
TOTAL SELLING EXPENSES	$ 75.00	$ 111.00	$ 147.00	$ 183.00	$ 188.00
GENERAL AND ADMINISTRATION	2025	2026	2027	2028	2029
Salaries and Wages	$ 7.00	$ 8.00	$ 9.00	$ 10.00	$ 11.00
Employee Benefits	$ 3.00	$ 4.00	$ 5.00	$ 6.00	$ 6.00
Payroll Taxes	$ 1.00	$ 2.00	$ 3.00	$ 4.00	$ 4.00
Insurance	$ 5.00	$ 6.00	$ 7.00	$ 8.00	$ 8.00
Rent	$ 7.00	$ 8.00	$ 9.00	$ 10.00	$ 10.00
Utilities	$ 2.00	$ 3.00	$ 4.00	$ 5.00	$ 5.00
Depreciation and Amortization	$ 3.00	$ 4.00	$ 5.00	$ 6.00	$ 6.00
Office Supplies	$ 1.00	$ 2.00	$ 3.00	$ 4.00	$ 4.00
Travel and Entertainment	$ 6.00	$ 7.00	$ 8.00	$ 9.00	$ 9.00
Postage	$ 1.00	$ 2.00	$ 3.00	$ 4.00	$ 4.00
Equipment Maintenance and Rental	$ 2.00	$ 3.00	$ 4.00	$ 5.00	$ 5.00
Interest	$ 1.00	$ 2.00	$ 3.00	$ 4.00	$ 4.00
Furniture and Equipment	$ 4.00	$ 5.00	$ 6.00	$ 7.00	$ 7.00
TOTAL GENERAL AND ADMINISTRATION EXPENSES	$ 43.00	$ 56.00	$ 69.00	$ 82.00	$ 83.00

TOTAL OPERATING EXPENSES	$ 118.00	$ 167.00	$ 216.00	$ 265.00	$ 271.00
NET INCOME BEFORE TAXES	$ (128.00)	$ (57.00)	$ 14.00	$ 85.00	$ 157.00
Taxes on Income	0	0	$ 2.80	$ 17.00	$ 31.40
NET INCOME AFTER TAXES	$ (128.00)	$ (57.00)	$ 11.20	$ 68.00	$ 125.60
Extraordinary Gain or Loss	$ 19.00	$ -	$ 10.00	$ -	$ -
Income Tax on Extraordinary Gain	$ 3.00	$ -	$ 2.00	$ -	$ -
NET INCOME (LOSS)	$ (112.00)	$ (57.00)	$ 19.20	$ 68.00	$ 125.60

LET'S PROCEED TO THE ACTIVITY!

THE USE OF PERSONAL FINANCE IN COMPUTER MODELLING

- It utilizes digital tools and software to manage, analyze, and plan financial activities like **budgeting, saving, debt repayment, and investments.** It provides users with visual insights, forecasts, and automated calculations, making financial planning more efficient and reliable.



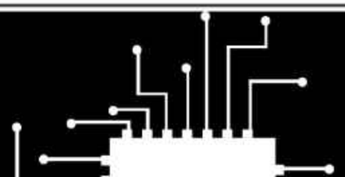
COMPARISON: COMPUTER MODELLING VS. PAPER-BASED & CALCULATOR METHODS

Aspect	Computer Modelling	Paper-Based and Calculator Methods
Accuracy	High, with automated calculations reducing errors.	Prone to human error in manual calculations.
Time Efficiency	Fast, with instant updates and simulations.	Slower, especially for complex calculations.
Data Storage	Stores and organizes data digitally.	Requires physical storage for paper records.
Scenario Simulation	Allows multiple financial scenarios and future forecasts.	Limited to static calculations without visual outputs.
Accessibility	Can be accessed on multiple devices (PC, phone, etc.).	Requires physical presence and tools like calculators.



ADVANTAGES

Advantages	Explanation
Automation	Performs complex calculations automatically.
Visualization	Provides graphs and charts for better understanding.
Scenario Analysis	Simulates "what-if" scenarios for better decision-making.
Time-Saving	Processes financial data quickly and efficiently.
Data Organization	Keeps records systematically for easy access and analysis.

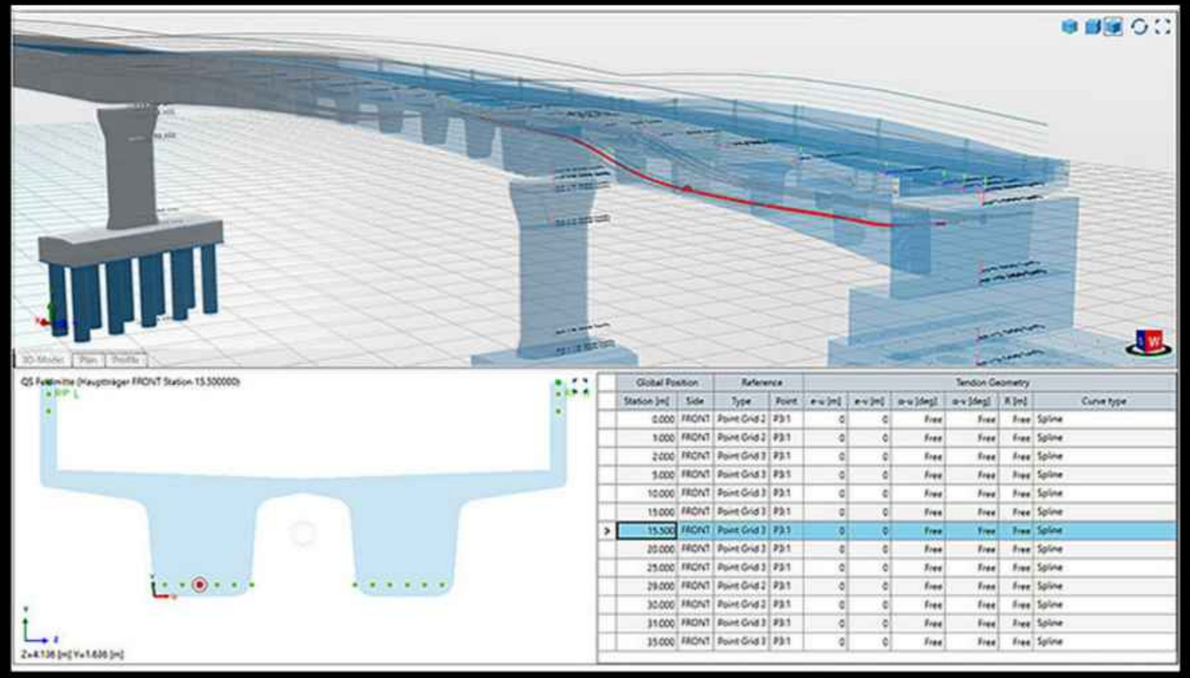


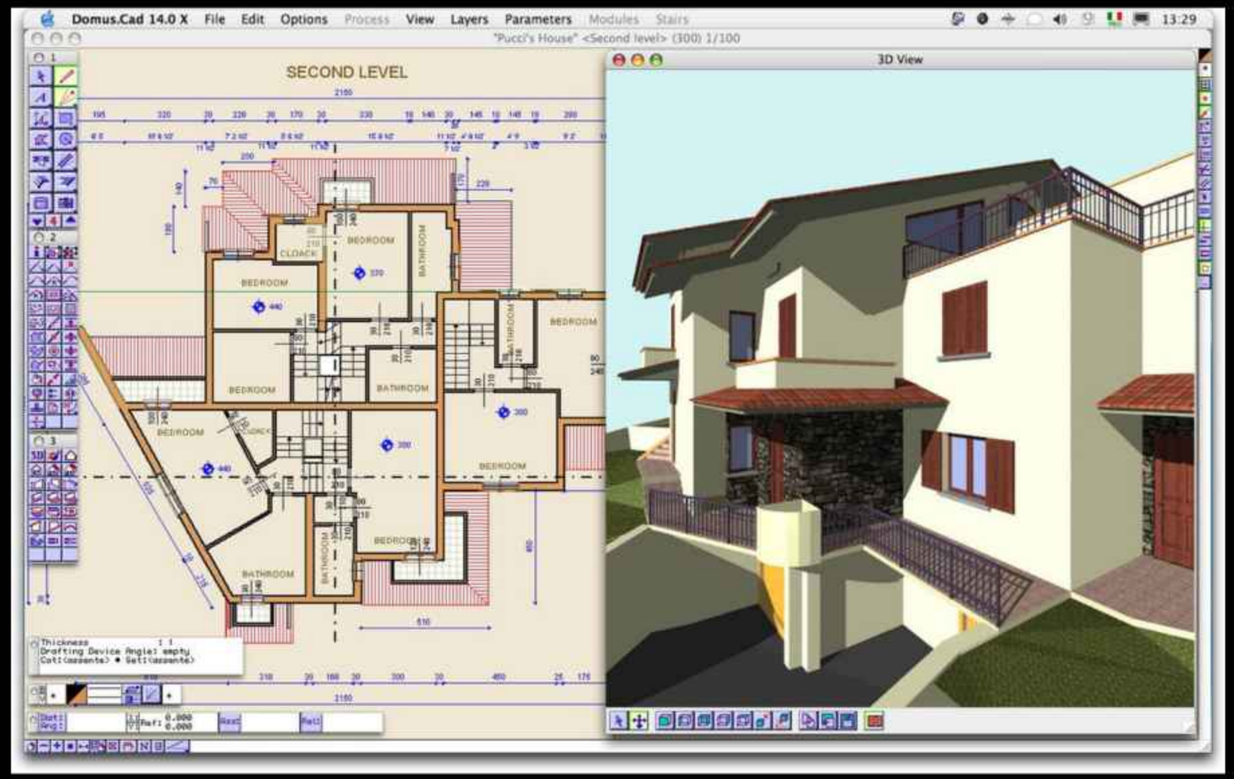
DISADVANTAGES

Disadvantages	Explanation
Dependence on Technology	Requires software, devices, and internet access.
Learning Curve	Users may need time to understand and use the software.
Costs	Some advanced tools or apps may have subscription fees.
Data Privacy Concerns	Risks of data breaches or unauthorized access.



THE USE OF COMPUTER MODELLING IN BRIDGE AND BUILDING DESIGN






blender


AutoCAD


SketchUp


AUTOCAD
CIVIL 3D

THE USE OF COMPUTER MODELLING IN BRIDGE AND BUILDING DESIGN

- It involves creating virtual representations of structures to simulate and analyze their performance under various conditions before construction. This approach is vital for ensuring safety, efficiency, and cost-effectiveness.



VIDEO PRESENTATION OF MODELLING APPLICATION



COMPARISON: COMPARISON OF COMPUTER MODELLING VS. REAL BRIDGE AND BUILDING DESIGN

Aspect	Computer Modelling	Real Bridge and Building Design
Process	Simulates designs digitally.	Involves physical construction and testing.
Time Efficiency	Faster due to automated calculations and simulations.	Slower as it requires physical prototypes and tests.
Cost	Lower, as no materials are wasted in initial testing.	Higher due to material costs in testing and revisions.
Safety	Identifies issues before actual construction.	Risks emerge during or after construction.
Environmental Impact	Evaluates impacts virtually, minimizing environmental risks.	May have unforeseen impacts detected after construction.



ADVANTAGES

Advantages	Explanation
Cost Savings	Reduces costs by eliminating the need for physical prototypes.
Time Efficiency	Speeds up the design process with automated tools.
Enhanced Safety	Simulates stress and natural disasters to prevent failures.
Improved Accuracy	Provides precise calculations for optimal design.
Sustainability	Analyzes environmental impact and promotes eco-friendly designs.



DISADVANTAGES

Disadvantages	Explanation
High Initial Cost	Software and expertise can be expensive.
Dependence on Data Accuracy	Errors in input data can lead to incorrect results.
Requires Skilled Professionals	Needs trained engineers and software specialists.
Limited to Simulations	Cannot fully replicate real-world conditions.

